

# Tharun Sridhar

Backend Engineer | Bengaluru, India

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## SUMMARY

Backend engineer (B.Tech CSE, VIT '26, CGPA 8.0/10) building production-style REST APIs in Django REST Framework and FastAPI on PostgreSQL, with JWT auth, Redis caching, Celery workers, Docker, and GitHub Actions CI/CD. Built the same inventory-management system in both stacks to benchmark the trade-offs firsthand.

## TECHNICAL SKILLS

**Languages:** Python, SQL, Java

**Backend:** Django, Django REST Framework, FastAPI, REST API Design, JWT Authentication, SQLAlchemy 2.0, Alembic, Celery, pytest

**Databases:** PostgreSQL 16, MySQL, Data Modeling, Indexing, Transaction Isolation, Row-Level Locking

**DevOps & Cloud:** Docker, Docker Compose, GitHub Actions (CI/CD), Redis, Gunicorn, Railway, AWS, Git, Linux

**AI / ML:** PyTorch, TensorFlow, Scikit-learn, LLM Integration, OpenCV, NumPy

**Frontend:** React.js, HTML, CSS

## EDUCATION

**Vellore Institute of Technology**

Vellore, India

*Bachelor of Technology in Computer Science | CGPA: 8.0/10*

*May 2026*

## PROJECTS

**Inventra** (*Team Lead*) | *Django, DRF, FastAPI, PostgreSQL 16, SQLAlchemy 2.0, Alembic, JWT, Docker* [GitHub](#)

- Led a team building a role-based inventory backend (Admin/Manager/Employee); personally built JWT auth (access + revocable refresh tokens) and the REST API over a versioned, indexed PostgreSQL 16 schema.
- Made stock mutations atomic and idempotent (append-only ledger) guarded by `SELECT ... FOR UPDATE` row-level locking in fixed order; regression-tested against a real concurrency race.
- Containerized with Docker and wired GitHub Actions CI to run the full pytest suite against PostgreSQL on every push; the API surface spans 51 endpoints across 13 modules with search, filtering, and pagination. Reimplemented the identical spec in FastAPI + SQLAlchemy 2.0 to benchmark the two stacks head-to-head.

**PhotoShare API** | *Django, DRF, FastAPI, PostgreSQL, Redis, Celery, JWT, ImageKit CDN, Docker* [GitHub](#)

- Built a photo/video sharing API with a full auth lifecycle (registration, verification, forgot/reset-password); reset tokens bind to the password hash so old tokens invalidate on change.
- Streamed multipart uploads to ImageKit CDN, added Redis cache-aside caching on feed/post-detail reads with write invalidation, and moved thumbnail generation to a retry-safe Celery worker with backoff.
- Enforced ownership-based authorization (403 on non-owner deletes); packaged API, worker, Postgres, and Redis via Docker Compose, with a parallel Django + DRF implementation of the same routes.

**NeuroScan AI** (*Team Lead*) | *Python, FastAPI, TensorFlow, OpenCV, Groq LLM* [GitHub](#)

- Directed a team on a brain-tumor MRI pipeline; personally trained a four-model classification ensemble and built the FastAPI serving layer, including an Attention U-Net stage reaching a 0.88 Dice score.
- Integrated Groq LLM report generation and PDF export via FastAPI, gating results into Accepted/Caution/Specialist-Review tiers using a Grad-CAM-vs-segmentation reliability index, covered by four pytest suites.

**Finvoro** | *Django, DRF, PostgreSQL, JWT, pytest* [GitHub](#)

- Built a personal finance API (Django REST Framework) for multi-account tracking, categorized transactions, monthly budgets, and spending reports, with balances computed live rather than cached.
- Enforced per-user data ownership on every endpoint via JWT (django-rest-framework-simplejwt), versioned under `/api/v1/`, and covered with a pytest + pytest-django suite plus Swagger/ReDoc docs.

## CERTIFICATIONS

**AWS Solutions Architect Associate (SAA-C03) – Exam Prep Course**

Packt (Coursera)

## ACHIEVEMENTS

5★ HackerRank: Problem Solving, Java, Python, SQL | [hackerrank.com/profile/tharunsridhar](https://hackerrank.com/profile/tharunsridhar)

162 LeetCode problems solved (73.75% acceptance) | [leetcode.com/u/Tharunsridhar](https://leetcode.com/u/Tharunsridhar)